Write here, write now! – Spelling difficulty disrupts parallel sentence planning.

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Poster: rpubs.com/jensroes/hsp-2026 (or use QR code)

Introduction

- Expressing ideas in language involves a cascade of processes starting with an intended message, moving on to retrieving words, their spelling or phonology, and finally outputting them in speech, onto paper or screen.
- Our understanding of the timecourse of linguistic planning and spelling retrieval in writing is limited.
- Upcoming linguistic information may be planned serially at linguistic edges (Flower and Hayes, 1980) or in parallel to production (Roeser et al., 2025).
- **We tested to what extent difficulty with orthography delays planning of upcoming information.**
- Importantly, only the parallel view predicts that disruptions delay planning of upcoming information.

Conclusion

- **Spelling difficulty disrupts parallel planning!** When writing ***the N1 and N2 ...*** N2 is retrieved while spelling N1.
- Hesitations related to N1 spelling affect N2 retrieval when there was not enough time for parallel planning.
- Written sentence planning unfolds in parallel to production: difficulty with spelling creates a planning bottleneck with knock-on effects on higher level planning.

Methods

Design

Task: Describe the image arrangement from left to right.

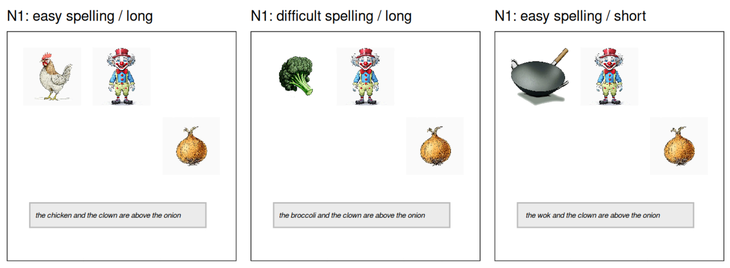

Figure 1: Example item (see e.g. Roeser et al., 2019).

Design: **N1 type** – see Fig. 1 – where **long** is ≥ 2 syllables and **difficult** is $H_{\text{spell}} > 0.5$. DV: inter-key intervals.

Prediction: Retrieval of N2 (*clown*) is delayed if N1 doesn't allow for parallel planning when N1 is difficult to spell (*broccoli*) or too short (*wok*) .

Spelling diversity

Images were based on Rossion and Pourtois (2004). Spelling diversity was defined as

$$H_{\text{spell}} = \sum_{i=1}^{K} p_i \times \log_2\left(\frac{1}{p_i}\right),$$

where k is the number of spellings for a picture name and p_i is the proportion of participants producing the ith spelling (Torrance et al., 2018).

Table 1: Example summary of "raccoon" with naming responses of 97 participants.

response	N_{resp}	N_{name}	Pr_{name}	H_{name}	Pr_{spell}	H_{spell}
raccoon	41	75	0.77	2.48	0.55	1.31
racoon	30	75	0.77	2.48	0.40	1.31
raccon	2	75	0.77	2.48	0.03	1.31
raccoo	1	75	0.77	2.48	0.01	1.31
racoo	1	75	0.77	2.48	0.01	1.31

Materials, Procedure & Participants

Thirty-six items with auditory prime (simple, conjoined subject NP) controlling for N1 retrieval; see Fig. 2.

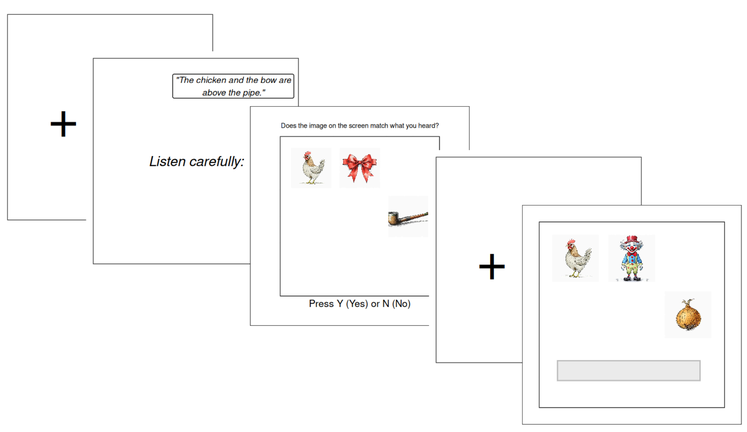

Figure 2: Example window sequence.

- 24 fillers with structures different form items.
- Sentence-picture match: YES for items; NO for fillers.

Ten ppts per counterbalancing list. Exp. 1: 56 ppts (6 lists), Exp. 2: 80 ppts (9 lists), Exp. 3: 96 ppts (12 lists).

Writing timecourse analysis

Hesitation probability θ was inferred from inter-key intervals (`iki`) using Bayesian mixed-effects mixture models (Roeser et al., 2025, 2024);

$$\left(\log\left(\text{iki}_{\text{ij}}\right) - \text{shift}_{\text{keyloc[i]}}\right) \sim \theta_{\text{keyloc,n1type}} \times \mathcal{N}\left(\beta_{\text{keyloc}} + \delta_{\text{keyloc}} + u_i + w_j, \sigma_{e'}^2\right) + \left(1 - \theta_{\text{keyloc,n1type}}\right) \times \mathcal{N}\left(\beta_{\text{keyloc}} + u_i + w_j, \sigma_e^2\right)$$

also illustrated in Fig. 3.

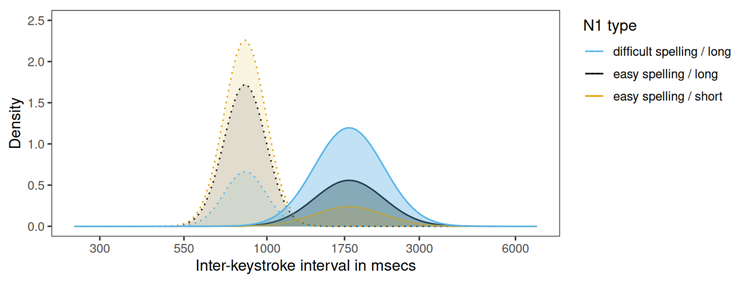

Figure 3: Hypothetical mixture of fluent (faded) and hesitant (solid) inter-key intervals.

Tutorial: rpubs.com/jensroes/mixture-models-tutorial

Experiment 1

Finding: Spelling difficulty increases hesitations before, during, and immediately **after** typing N1.

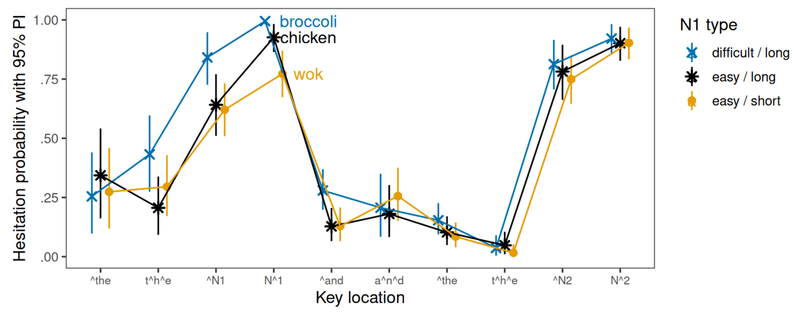

Figure 4: Posterior hesitation probability with 95% probability intervals (PIs).

Experiment 2

Prime without N2 determiner: ***The N1 and N2*** *are above the N3* (vs *The N1 and the N2 . . .*)

Finding: When N1 does not allow parallel planning, *"the"*-omission increases post-N1 hesitations.

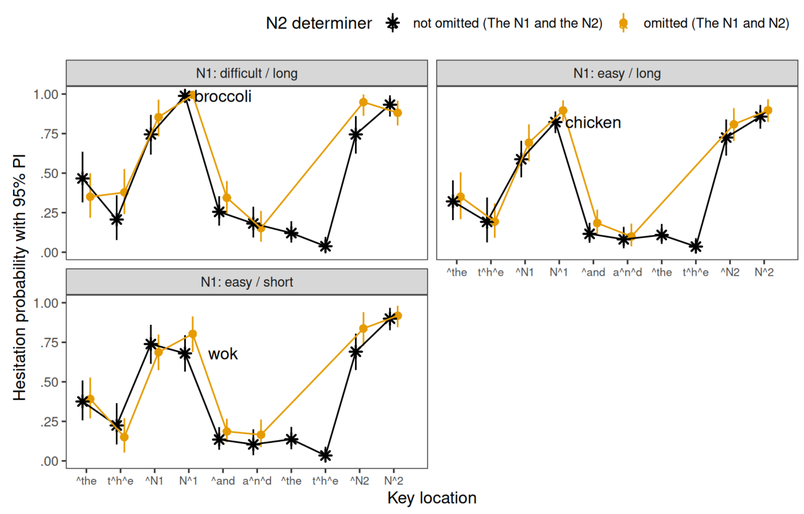

Figure 5: Posterior hesitation probability with 95% probability intervals (PIs).

Experiment 3

(Auditory) prime with N2 preview: ***The N1 and clown*** *are above the pipe* (vs *The N1 and bow*)

Finding: Post-N1 hesitations are associated with N2 retrival.

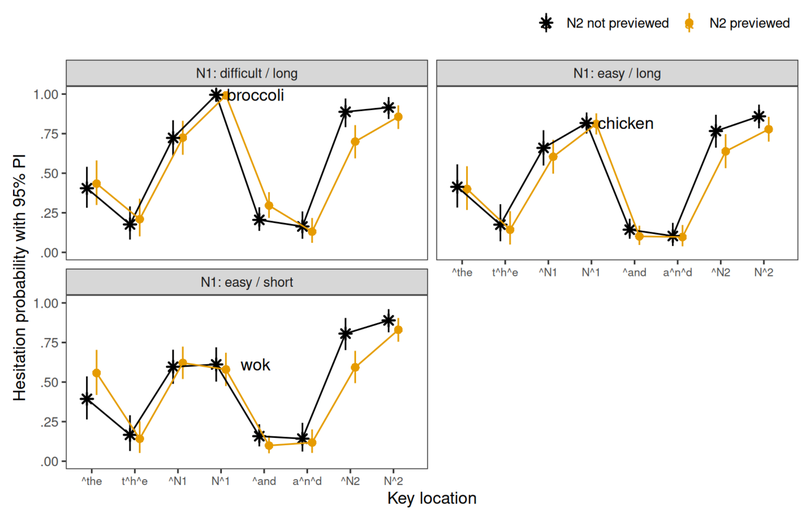

Figure 6: Posterior hesitation probability with 95% probability intervals (PIs).

References

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